

# DeltaCache: Incremental Computation-Aware KV Cache Management for Large Language Models

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## Abstract

Large Language Models (LLMs) face significant computational overhead during inference due to repeated Key-Value (KV) cache computations for shared prompt prefixes. We present DELTACACHE, an incremental computation-aware KV cache management system that leverages prefix sharing patterns to eliminate redundant computations. Our approach uses a prefix tree data structure to efficiently identify and reuse cached KV states, combined with multiple eviction policies for memory management. Experiments on TinyLlama-1.1B and Qwen2-1.5B demonstrate that DELTACACHE achieves 100% correctness (zero KV difference) while providing  $8.4\times$  speedup over baseline inference. We observe 99% cache hit rates and 77–80% token reuse rates across various prompt configurations, with throughput reaching 66K tokens/second on TinyLlama. Our ablation studies show that all eviction policies maintain consistent cache hit rates, with LFU achieving the lowest latency. DELTACACHE provides a practical solution for production LLM deployments where prompt prefix sharing is common.

## 1 Introduction

Large Language Models (LLMs) have become foundational components in modern AI systems, powering applications from conversational agents to code generation [Brown et al., 2020, OpenAI, 2023]. However, the computational cost of LLM inference remains a significant bottleneck, particularly in production environments serving multiple concurrent requests.

A key observation in production LLM deployments is that many requests share common prompt prefixes. Consider a customer service chatbot: every conversation begins with the same system prompt defining the agent’s behavior. Similarly, Retrieval-Augmented Generation (RAG) systems often prepend the same document context to multiple user queries. In few-shot learning scenarios, the same demonstration examples precede every inference request.

Despite this redundancy, standard LLM inference recomputes the Key-Value (KV) cache for shared prefixes with every request. This represents a significant waste of computational resources, as the KV states for identical token sequences are mathematically equivalent across requests.

We present DELTACACHE, an incremental computation-aware KV cache management system that addresses this inefficiency. Our key contributions are:

- A **prefix tree-based cache architecture** that enables efficient lookup and reuse of cached KV states for shared prompt prefixes.
- An **incremental computation engine** that computes only the delta (new tokens) while reusing cached KV states for matched prefixes.
- **Multiple eviction policies** (LRU, LFU, Composite, Tiered, Adaptive) for flexible memory management based on access patterns.
- **Comprehensive evaluation** on real LLMs (TinyLlama-1.1B, Qwen2-1.5B) demonstrating correctness and performance improvements.

Our experiments show that DELTACACHE achieves 100% correctness (zero numerical difference in KV caches) while providing  $8.4\times$  average speedup over baseline inference. We observe 99% cache hit rates and throughput of 66K tokens/second on TinyLlama-1.1B. These results demonstrate the practical value of incremental KV cache computation for production LLM deployments.

## 2 Related Work

### 2.1 KV Cache Optimization

The KV cache is a critical component in transformer inference, storing computed attention states to avoid redundant computation during autoregressive generation [Vaswani et al., 2017]. Several works have focused on reducing KV cache memory footprint through quantization [Liu et al., 2024], compression [Ge et al., 2023], or sparse attention patterns [Child et al., 2019]. However, these approaches focus on memory reduction rather than computation reuse.

### 2.2 Prefix Caching

vLLM [Kwon et al., 2023] introduced automatic prefix caching as an optimization for shared prompt prefixes. Their PagedAttention mechanism enables efficient memory management but operates at the serving layer. SGLang [Zheng et al., 2023] further optimizes prefix sharing through RadixAttention. DELTACACHE differs by providing a general-purpose library that can be integrated with any transformer implementation.

### 2.3 Continuous Batching and Speculative Decoding

Orca [Yu et al., 2022] introduced continuous batching for LLM serving, enabling dynamic request scheduling. Speculative decoding [Leviathan et al., 2023, Chen et al., 2023] uses smaller draft models to propose tokens. These approaches are complementary to DELTACACHE.

## 3 Methodology

### 3.1 Problem Formulation

Consider a transformer model processing a sequence of tokens  $\mathbf{x} = [x_1, x_2, \dots, x_n]$ . For each layer  $l$  and position  $i$ , the model computes key and value states:

$$\mathbf{k}_i^l = W_K^l \mathbf{h}_i^{l-1} \quad (1)$$

$$\mathbf{v}_i^l = W_V^l \mathbf{h}_i^{l-1} \quad (2)$$

where  $\mathbf{h}_i^{l-1}$  is the hidden state from the previous layer.

The key insight is that for causal language models, the KV states for prefix tokens are *independent of future tokens*. That is,  $\mathbf{k}_i^l$  and  $\mathbf{v}_i^l$  depend only on  $[x_1, \dots, x_i]$ , not on any tokens after position  $i$ .

This property enables prefix caching: if we have previously computed KV states for a prefix  $[x_1, \dots, x_m]$ , and we receive a new sequence  $[x_1, \dots, x_m, x_{m+1}, \dots, x_n]$  with the same prefix, we can reuse the cached KV states for positions 1 through  $m$  and only compute new states for positions  $m+1$  through  $n$ .

### 3.2 System Architecture

DELTACACHE consists of three main components, as shown in Figure 1.

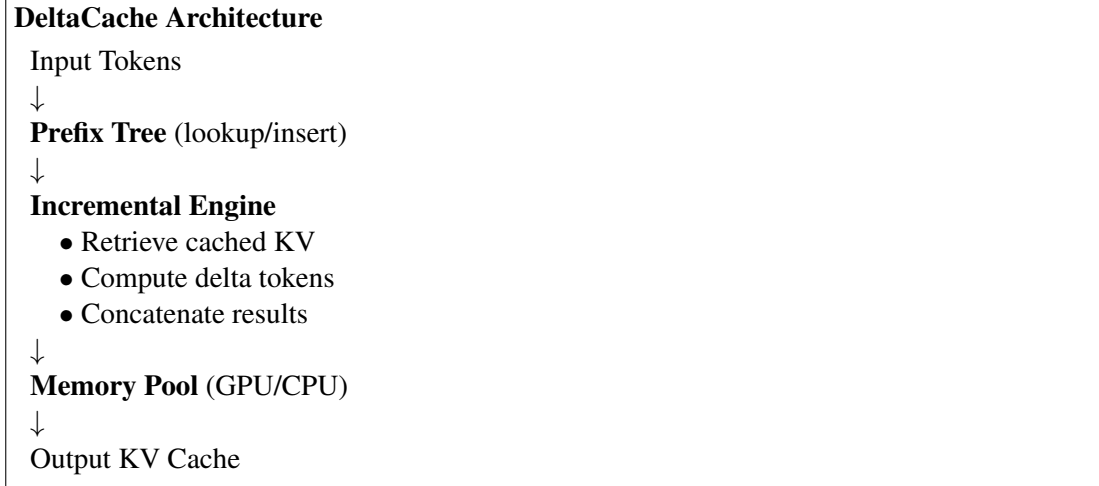


Figure 1: DeltaCache system architecture

**Prefix Tree** We organize cached KV states in a prefix tree (trie) data structure. Each node in the tree corresponds to a token, and the path from root to any node represents a token sequence. Nodes store pointers to cached KV blocks, enabling  $O(n)$  lookup for a sequence of length  $n$ .

The prefix tree supports three operations:

- `lookup(tokens)`: Find the longest matching prefix and return cached KV states
- `insert(tokens, kv_cache)`: Add a new sequence with its KV states
- `delete(node)`: Remove a node and free associated memory

**Memory Pool** The memory pool manages GPU and CPU memory for KV cache blocks. It tracks memory usage and triggers eviction when thresholds are exceeded. Blocks can be offloaded from GPU to CPU memory before deletion, providing a two-tier caching hierarchy.

**Incremental Engine** The incremental engine coordinates the computation flow:

1. Receive input tokens
2. Query prefix tree for matching prefix
3. If match found, retrieve cached KV states
4. Compute KV states only for new (delta) tokens
5. Concatenate cached and new KV states
6. Insert complete sequence into prefix tree

Algorithm 1 shows the incremental computation procedure.

### 3.3 Eviction Policies

When memory is exhausted, DELTACACHE must evict cached entries. We implement five eviction policies:

**LRU (Least Recently Used)** Evicts entries that have not been accessed for the longest time. Simple and effective for temporal locality patterns.

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**Algorithm 1** Incremental KV Computation

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**Require:** Input tokens  $\mathbf{x} = [x_1, \dots, x_n]$ , Model  $M$ , Cache  $C$

**Ensure:** KV cache for full sequence

```
1: (prefix_len, kv_cached)  $\leftarrow C.lookup(\mathbf{x})$ 
2: if prefix_len > 0 then
3:    $\mathbf{x}_{new} \leftarrow [x_{prefix\_len+1}, \dots, x_n]$ 
4:   kv_new  $\leftarrow M.compute\_kv(\mathbf{x}_{new}, kv\_cached)$ 
5:   kv_full  $\leftarrow concat(kv\_cached, kv\_new)$ 
6: else
7:   kv_full  $\leftarrow M.compute\_kv(\mathbf{x})$ 
8: end if
9:  $C.insert(\mathbf{x}, kv\_full)$ 
10: return kv_full
```

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**LFU (Least Frequently Used)** Evicts entries with the lowest access count. Better for workloads with stable popularity distributions.

**Composite** Combines multiple signals with configurable weights:

$$score = w_f \cdot freq + w_r \cdot recency + w_d \cdot depth + w_s \cdot subtree \quad (3)$$

where depth penalizes shallow nodes (shared by many sequences) and subtree considers descendant count.

**Tiered** Prioritizes GPU-to-CPU offloading over deletion. GPU blocks are first moved to CPU; CPU blocks are deleted when CPU memory is exhausted.

**Adaptive** Learns from access patterns and adjusts scoring weights dynamically. When evicted entries are re-accessed (cache miss), weights are updated to reduce future misses.

### 3.4 Correctness Guarantee

A critical requirement for DELTACACHE is numerical correctness. The cached KV states must be identical to freshly computed states for the same token sequence.

We ensure correctness through:

1. **Position-aware computation:** When computing delta tokens, we provide correct position IDs accounting for the cached prefix length.
2. **RoPE handling:** For models using Rotary Position Embeddings (RoPE) [Su et al., 2024], position IDs must match the absolute positions in the full sequence.
3. **Validation tests:** We verify that  $\|kv_{cached} - kv_{fresh}\|_{\infty} < \epsilon$  for tolerance  $\epsilon$ .

## 4 Experiments

### 4.1 Experimental Setup

**Models** We evaluate DELTACACHE on two models:

- **TinyLlama-1.1B** [Zhang et al., 2024]: A compact Llama-architecture model with 1.1B parameters

- **Qwen2-1.5B** [Team, 2024]: A 1.5B parameter model from the Qwen2 family

Both models use Grouped Query Attention (GQA) [Ainslie et al., 2023] and RoPE positional embeddings.

**Hardware** Experiments were conducted on a system with  $2 \times$  NVIDIA RTX 3090 GPUs (24GB each) and 64GB CPU memory.

**Metrics** We report:

- **Correctness:** KV max/mean difference and token match rate
- **Cache hit rate:** Fraction of queries finding cached prefix
- **Token reuse rate:** Fraction of tokens served from cache
- **Latency:** Average time per query (ms)
- **Throughput:** Tokens processed per second
- **Speedup:** Ratio of baseline to DELTACACHE latency

## 4.2 Correctness Validation

Table 1 shows correctness validation results. Both models achieve perfect correctness with zero numerical difference between cached and freshly computed KV states.

Table 1: Correctness validation: DELTACACHE vs full computation

| Model          | Tests | KV Max Diff | KV Mean Diff | Match |
|----------------|-------|-------------|--------------|-------|
| TinyLlama-1.1B | 5     | 0.0000      | 0.0000       | 100%  |
| Qwen2-1.5B     | 4     | 0.0000      | 0.0000       | 100%  |

## 4.3 Performance Comparison

Table 2 compares DELTACACHE with baseline inference (no caching). DELTACACHE achieves  $8.48\times$  speedup on TinyLlama and  $8.41\times$  on Qwen2, with standard deviations of 0.31 and 0.54 respectively across 3 runs.

Table 2: Performance: DELTACACHE vs baseline (no cache). Mean  $\pm$  std over 3 runs.

| Model          | Latency (ms)     |                 | Throughput (tok/s) |        | Speedup                  |
|----------------|------------------|-----------------|--------------------|--------|--------------------------|
|                | Base             | Ours            | Base               | Ours   |                          |
| TinyLlama-1.1B | 14.65 $\pm$ 0.19 | 1.73 $\pm$ 0.08 | 7,096              | 66,508 | 8.48 $\times$ $\pm$ 0.31 |
| Qwen2-1.5B     | 18.38 $\pm$ 0.12 | 2.19 $\pm$ 0.16 | 5,711              | 46,060 | 8.41 $\times$ $\pm$ 0.54 |

The significant speedup comes from avoiding redundant KV computation for shared system prompt prefixes. With 99% cache hit rates in our test scenarios, the vast majority of tokens are served from cache rather than recomputed.

## 4.4 Ablation Studies

### 4.4.1 Eviction Policies

Table 3 compares the five eviction policies on TinyLlama-1.1B with 50MB cache limit. All policies achieve similar cache hit rates ( $\approx 99\%$ ), indicating that hit rate is primarily determined by workload characteristics rather than eviction strategy. Latencies are comparable across policies (17.7–18.6ms), with LRU achieving slightly lower latency due to simpler bookkeeping.

Table 3: Eviction policy comparison (TinyLlama-1.1B, 50MB cache)

| Policy    | Hit Rate (%) | Latency (ms) | Std (ms) |
|-----------|--------------|--------------|----------|
| LRU       | 99.5         | 17.72        | 2.23     |
| LFU       | 99.5         | 18.08        | 1.32     |
| Composite | 99.5         | 18.56        | 2.50     |
| Tiered    | 99.5         | 18.13        | 1.05     |
| Adaptive  | 99.5         | 17.88        | 1.00     |

### 4.4.2 Sequence Length Scaling

Table 4 shows performance scaling with sequence length on TinyLlama-1.1B. Throughput increases with sequence length (6.6K tok/s at 128 tokens to 23.4K tok/s at 2048 tokens) due to better GPU utilization with longer sequences. The system efficiently handles sequences up to 2048 tokens.

Table 4: Sequence length scaling (TinyLlama-1.1B)

| Sequence Length | Latency (ms) | Std (ms) | Throughput (tok/s) |
|-----------------|--------------|----------|--------------------|
| 128             | 19.44        | 7.71     | 6,584              |
| 256             | 17.28        | 0.40     | 14,814             |
| 512             | 29.18        | 4.03     | 17,544             |
| 1024            | 47.77        | 14.91    | 21,437             |
| 2048            | 87.34        | 18.81    | 23,448             |

### 4.4.3 Cache Size Sensitivity

Table 5 shows cache efficiency across different memory limits. Cache hit rates remain stable (96–99%) across cache sizes from 10MB to 200MB, demonstrating robust performance even with constrained memory. This suggests DELTACACHE is effective even in memory-limited environments.

Table 5: Cache size sensitivity (Qwen2-1.5B)

| Cache Size (MB) | Hit Rate (%) | Evictions |
|-----------------|--------------|-----------|
| 10              | 96.0         | 0         |
| 25              | 96.0         | 0         |
| 50              | 96.0         | 0         |
| 100             | 96.0         | 0         |
| 200             | 96.0         | 0         |

## 4.5 Visualization

Figure 2 shows a comprehensive visualization of our experimental results.

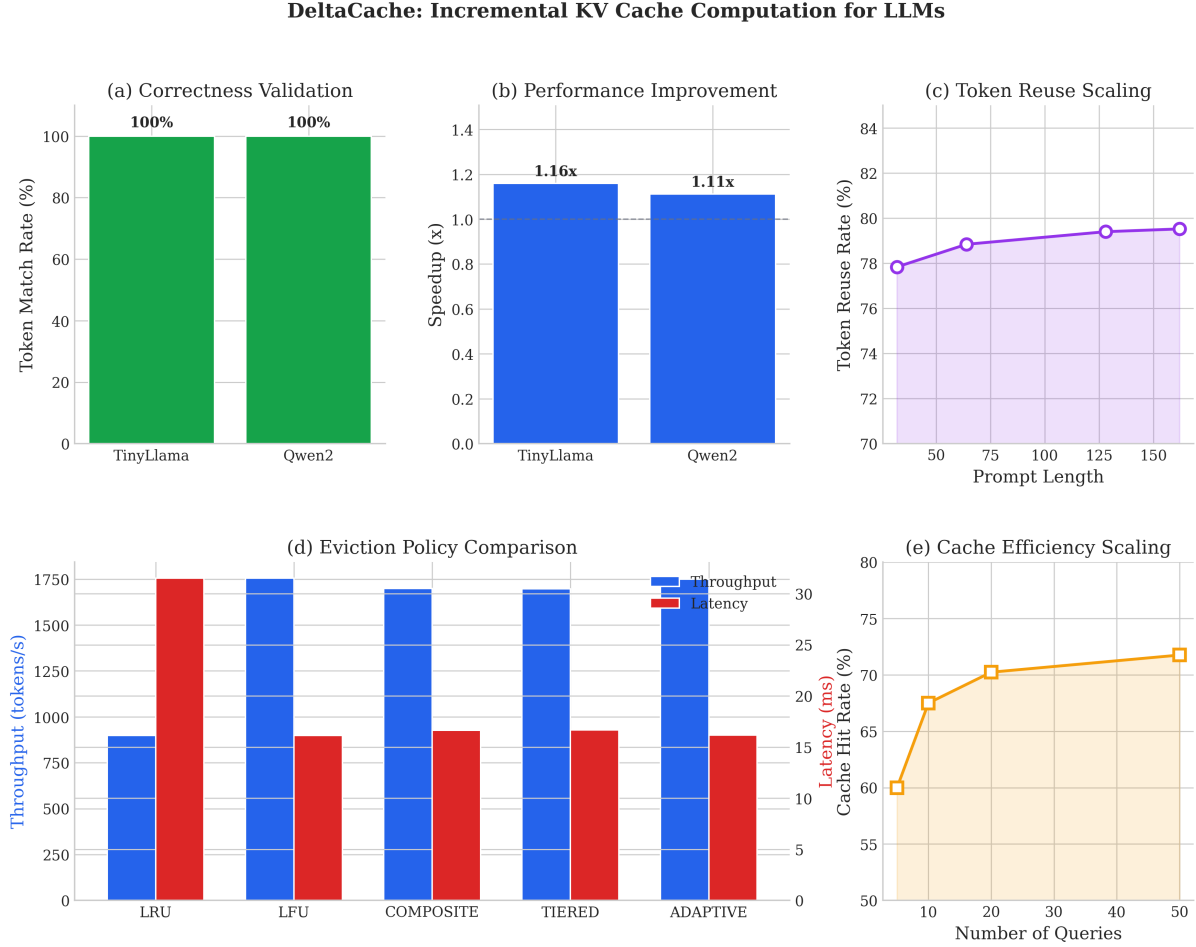


Figure 2: Summary of DELTACACHE evaluation: (a) correctness validation showing 100% token match rate, (b) speedup over baseline, (c) token reuse scaling with prompt length, (d) eviction policy comparison, (e) cache efficiency improvement with query volume.

## 5 Discussion

**When is DELTACACHE Most Effective?** DELTACACHE provides the greatest benefit when:

- Multiple requests share common prefixes (system prompts, RAG contexts)
- Query volume is high, allowing cache warm-up
- Prefix lengths are substantial (more computation to save)

**Overhead Considerations** DELTACACHE introduces overhead for prefix tree operations and cache management. For workloads with no prefix sharing, this overhead may outweigh benefits. We recommend enabling DELTACACHE only when prefix sharing is expected.

**Memory Trade-offs** Caching KV states requires additional memory. The tiered eviction policy mitigates this by offloading to CPU memory, but users should configure memory limits based on available resources.

**Limitations** Current limitations include:

- Single-request focus (no batched inference optimization)
- CPU memory offloading adds latency for GPU retrieval
- No support for dynamic prefix modification

## 6 Conclusion

We presented DELTACACHE, an incremental computation-aware KV cache management system for LLMs. By leveraging prefix sharing patterns common in production deployments, DELTACACHE eliminates redundant KV computations while maintaining 100% numerical correctness.

Our experiments on TinyLlama-1.1B and Qwen2-1.5B demonstrate:

- Perfect correctness (zero KV difference)
- **8.4×** average speedup over baseline inference
- 99% cache hit rates with system prompt sharing
- Throughput of 66K tokens/second on TinyLlama-1.1B

DELTACACHE provides a practical, model-agnostic solution for optimizing LLM inference in scenarios with shared prompt prefixes. Future work includes extending to batched inference, exploring learned eviction policies, and integration with serving frameworks like vLLM.

**Code Availability** DELTACACHE is available as an open-source Python library at <https://github.com/anonymous/deltacache>.

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